

PCAP TRAFFIC ANALYSIS REPORT

Intrusion Detection — Network Forensics Investigation

CCP-CNS2026 · Computer Networks · Problem 6

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1. Executive Summary

This report presents a comprehensive forensic analysis of a 42 MB packet capture file (2025-06-13-traffic-analysis-exercise.pcap) provided as part of the CCP-CNS2026 Problem 6 Intrusion Detection exercise. The capture spans 34 minutes and contains 48,877 packets recorded between 20:33:55 and 21:08:28 on June 13, 2025.

Analysis using Wireshark 4.x identified a large-scale encrypted data exfiltration and Command-and-Control (C2) beaconing attack conducted over TLS 1.2 on TCP port 443. The external attacker IP 83.137.149.15 communicated with the internal victim host 10.6.13.133, exchanging 35,485 packets totalling approximately 34 MB — making it the single dominant conversation in the entire capture at 85.8% of all traffic.

The first suspicious packet was identified as Packet #8567 at timestamp 157.661094s, a SYN,ACK from the attacker on port 443, marking the establishment of the malicious TLS session. The recommended immediate mitigation is to block all outbound connections from 10.6.13.133 to 83.137.149.15 on port 443 at the perimeter firewall.

1.1 Key Findings at a Glance

Field	Value
PCAP File	2025-06-13-traffic-analysis-exercise.pcap
File Size	42 MB · 48,877 packets · 34 minutes
Attacker IP	83.137.149.15 (external)
Victim IP	10.6.13.133 (internal)
Attack Type	Encrypted data exfiltration / C2 beaconing over TLS
Protocol	TLSv1.2 over TCP port 443 (85.8% of all traffic)
First Suspicious Packet	Packet #8567 at 157.661094s — SYN,ACK on port 443
Volume	35,485 packets / 34 MB (attacker-victim conversation)
Recommended Mitigation	Block 10.6.13.133 → 83.137.149.15 on port 443 at firewall

2. Analysis Methodology

The analysis followed a structured five-phase forensic workflow applied to the instructor-provided PCAP file. All analysis was performed using Wireshark 4.x on Windows. The PCAP file integrity was verified prior to analysis using the SHA-256 hash provided in Wireshark's Capture File Properties.

2.1 Phase 1 — File Verification & Baseline Statistics

The capture file was opened in Wireshark and its properties examined via Statistics → Capture File Properties. Key baseline metrics were recorded: 48,877 total packets, 42,106,050 bytes (42 MB), average packet size of 861 bytes, and a capture duration of 34 minutes 32 seconds. The SHA-256 hash was noted for evidence integrity documentation.

2.2 Phase 2 — Conversation Analysis (Attacker & Victim Identification)

Wireshark's Statistics → Conversations view (IPv4 tab) was used to identify the top talkers in the capture. The conversation list was sorted by packet count in descending order. The top entry immediately revealed an anomalous communication: 10.6.13.133 (internal host) exchanging 35,485 packets and 34 MB with the external IP 83.137.149.15. This volume was orders of magnitude higher than any other conversation in the capture, immediately flagging this pair as the attacker-victim hosts.

2.3 Phase 3 — Display Filter Application

Five Wireshark display filters were applied sequentially to isolate and examine specific traffic patterns:

Filter	Syntax	Purpose	Packets Displayed
Source IP Filter	ip.src == 83.137.149.15	Isolate all outbound traffic from attacker	23,281 (47.6%)
Destination IP Filter	ip.dst == 10.6.13.133	Isolate all inbound traffic to victim	30,219 (61.8%)
Port 443 Filter	tcp.port == 443	Confirm HTTPS/TLS as primary protocol	41,941 (85.8%)
SYN Flag Filter	tcp.flags.syn == 1	Identify all TCP connection initiations	426 (0.9%)
Conversations Stats	Statistics > Conversations	Rank all hosts by traffic volume	All 48,877 packets

2.4 Phase 4 — First Suspicious Packet Identification

With the filter `ip.src == 83.137.149.15` active, the packet list was examined from the earliest timestamp. Packet #8567 at timestamp 157.661094s was identified as the first packet originating from the attacker. This packet is a TCP SYN,ACK (Seq=0, Ack=1, Win=65535, MSS=1396, WS=256, SACK_PERM) from port 443 to victim port 52527, marking the server-side response in the TCP three-way handshake and thus the beginning of the malicious TLS session establishment.

Prior to this packet, at timestamp 157.513273s, a DNS query response was observed (Packet #8565) resolving `windows.php.net` to 83.137.149.15 — revealing that the attacker's infrastructure was masquerading under a domain name designed to appear legitimate. This DNS resolution preceded the TCP connection, consistent with a C2 beacon check-in sequence.

2.5 Phase 5 — Attack Classification & Pattern Analysis

Post-handshake traffic analysis revealed the following behavioral indicators examined to classify the attack type and assess its severity:

- Sustained high-volume data transfer: 34 MB over 34 minutes (approximately 1 MB/min average) flowing primarily from the attacker to the victim.
- TLS 1.2 encryption: The full TLS handshake was captured (Client Hello with SNI=windows.php.net, Server Hello, Certificate, Server Key Exchange, Change Cipher Spec) followed by sustained Application Data frames.
- Large TCP segment sizes: The majority of data packets are 1430 bytes (maximum segment size), indicating bulk data transfer optimized for throughput.
- DNS camouflage: The C2 server resolved via a domain (windows.php.net) designed to mimic legitimate Microsoft infrastructure, a common APT evasion technique.
- Port 443 abuse: Using HTTPS port 443 allows the malicious traffic to blend with normal web traffic and bypass most perimeter inspection rules.

3. Findings, Classification & Mitigation

3.1 Attack Classification

Based on the evidence gathered across all five analysis phases, the observed traffic is classified as:

Primary Classification: Encrypted Data Exfiltration via Command-and-Control (C2) Beaconing over TLS

This classification is justified by the following evidence:

- Volume asymmetry: The attacker-to-victim direction (33 MB inbound to victim) represents a data push into the victim, consistent with malware payload delivery or C2 command issuance, while victim-to-attacker (754 kB) represents beaconing acknowledgments.
- Persistence: The attack session lasted approximately 136 seconds (Relative Start: 157.51s, Duration: 136s) with continuous data flow rather than the bursty pattern typical of port scans or brute force.
- Single dominant session: 85.8% of all capture traffic belongs to this single conversation, ruling out distributed DoS (which produces many-to-one patterns) and port scans (which produce one-to-many patterns).
- TLS encryption: Encrypting C2 traffic over port 443 is a well-documented APT and malware technique to evade deep-packet inspection and blend with legitimate HTTPS business traffic.
- Domain impersonation: The windows.php.net domain used to resolve the C2 server is consistent with adversary infrastructure designed to evade DNS blocklists and appear as routine Microsoft update traffic.

3.2 Indicators of Compromise (IOCs)

IOC Type	Value	Significance
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Attacker IP	83.137.149.15	External C2 server; top talker in capture
Victim IP	10.6.13.133	Internal compromised host
C2 Domain	windows.php.net	Domain masquerading as Microsoft; resolved to attacker IP
TCP Port	443 (HTTPS)	Used to disguise C2 traffic as legitimate web traffic
TLS Version	TLSv1.2	Encrypted channel used for exfiltration / C2 commands
First Packet	#8567 at 157.661094s	Initial TCP SYN,ACK establishing malicious session
Data Volume	35,485 pkts / 34 MB	Anomalously high single-session volume

3.3 Mitigation Recommendations

The following mitigations are recommended in order of priority:

Immediate (Priority 1): Block all outbound connections from internal host 10.6.13.133 to external IP 83.137.149.15 on TCP port 443 at the perimeter firewall. Simultaneously isolate host 10.6.13.133 from the network for forensic imaging.

Short-term (Priority 2): Add 83.137.149.15 and the domain windows.php.net to the DNS sinkhole and IP blocklist. Enable TLS inspection (SSL/TLS decryption) on the perimeter firewall to detect future encrypted C2 channels.

Long-term (Priority 3): Deploy a Network Intrusion Detection System (NIDS) such as Snort or Zeek with rules tuned to detect anomalous outbound data volumes on port 443. Implement network behavioral analytics to flag single-session data volumes exceeding a defined threshold (e.g., >5 MB in <10 minutes).

3.4 Conclusion

The PCAP analysis conclusively identifies a targeted, encrypted C2 exfiltration attack against internal host 10.6.13.133 by external threat actor 83.137.149.15. The attacker's use of TLS encryption over port 443, combined with a legitimacy-mimicking domain name, represents a sophisticated evasion strategy consistent with Advanced Persistent Threat (APT) tradecraft.

The analysis demonstrates that even encrypted traffic can be identified as malicious through behavioral indicators: session volume, traffic asymmetry, connection duration, and DNS anomalies. Immediate firewall-level blocking of the identified attacker IP, combined with host isolation and a forensic investigation of 10.6.13.133, is essential to contain this compromise and prevent further data loss.

References

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